Engineering Journal

Team name:

Uniqa CRO team

RoboCup Junior

1th – 7 th of July

Incheon, South Korea

11th August 2025
Lovro Žigman

- Team was formed during the Robotics Summer School on Hvar.
- Lovro Žigman was selected as team captain.
- Team members agreed to begin independent research into the Rescue Simulation category.
- Rescue Simulation was chosen because it enabled efficient remote collaboration between team members located in different cities.

3rd December 2025
Orsat Kralj

Plan for today:

- Define team responsibilities.

Team roles:

- Lovro Žigman – Team captain, robot designer, navigation and path planning developer.
- Luka Kolarić – AI developer, camera integration, victim detection system developer, testing support.
- Orsat Kralj – Poster designer, simulation testing, software support and debugging.

4th December 2025
Lovro Žigman

Plan for today:

- Design robot architecture.

- Select sensors and define their purposes.

Selected sensors:

- 8 distance sensors
- 2 cameras
- GPS
- Inertial Unit (IMU)
- Colour sensor
- LiDAR

Initial design goals:

- Reliable autonomous navigation.
- Environment mapping.
- Victim and hazard detection.
- Efficient localization.

LiDAR was selected as the primary mapping sensor, while GPS and IMU would provide localization data.

10th December 2025
Luka Kolarić

Current challenges:

Cameras:

- Victim detection was unreliable.
- Decision: postpone implementation until navigation system becomes stable.

LiDAR:

- Inaccurate wall interpretation in some maze configurations.
- Solution: improve wall processing logic and map generation.

Mapping:

- Team still lacked experience with grid-based mapping systems.
- Orsat continued researching path planning algorithms and mapping techniques.

15th December 2025
Orsat Kralj

Research completed on navigation algorithms:

- A* Algorithm
- Dijkstra Algorithm

Conclusion:

- A* appeared more suitable because it provides faster route calculation toward known targets.
- Further investigation into Webots implementation was required.

20th December 2025
Lovro Žigman

Progress update:

- Hole detection using distance sensors was implemented successfully.

- LiDAR processing was improved.
- Robot became significantly more reliable when navigating narrow corridors.
- Early mapping functionality was integrated into the navigation system.

26th December 2025
Luka Kolarić

Progress update:

- Exit procedure after complete exploration was implemented.
- Team decided to adopt the A* algorithm as the primary path planning method.
- Initial implementation of grid-based navigation began.

12th January 2026
Lovro Žigman

Main objective:

- Improve navigation reliability before the national competition.

Focus areas:

- Complete maze exploration.
- Avoid dangerous areas.
- Improve route planning efficiency.
- Minimize navigation errors.

18th January 2026
Lovro Žigman

Testing results:

- Navigation system performed reliably in Erebus test environments.
- Mapping quality improved significantly.
- Additional test maps were required for further optimization.

22nd January 2026
Lovro Žigman

Progress update:

- Additional maps tested successfully.
- Path planning became more stable.
- Continued optimization of robot movement and localization.

12th March 2026
Luka Kolarić

Progress update:

- Navigation system reached stable performance.
- Work began on victim detection improvements and AI integration.

14th March 2026
Lovro Žigman

National competition results:

- Navigation performance was highly reliable.
- Robot successfully explored nearly all test environments.
- Mapping system performed consistently.
- Team achieved 2nd place at the Croatian National RoboCup Junior Competition.
- Qualification for the RoboCup Junior World Championship was secured.

16th May 2026
Lovro Žigman

Major milestone:

- Team Description Paper completed.
- Navigation and mapping systems finalized.
- AI victim detection development accelerated.

16th June 2026
Orsat Kralj

World Championship preparation:

- Competition poster completed.
- Extensive simulation testing performed.
- YOLO-based victim detection system integrated.
- Hazard detection using HSV colour analysis completed.
- Robot reached its most advanced version before the World Championship in Incheon.